

The Language of Electoral Competition: Thematic Structure and Ideological Framing in French Legislative Manifestos, 1973–1993

Gabriel Orsatti*

ENSAE Paris - Institut Polytechnique de Paris

M2, 2025–2026

`gabriel.orsatti@ensae.fr`

github: gabrielorsatti

Abstract

This paper applies natural language processing techniques to a corpus of 21,697 French electoral manifestos (*professions de foi*) spanning five legislative elections between 1973 and 1993. We combine unsupervised LDA topic modeling, supervised text classification (logistic regression on TF-IDF features), and log-odds rhetorical analysis to answer two fundamental questions: what underlying thematic structure organises political discourse, and how do distinct political families frame identical policy themes? An 11-topic LDA model produces a highly balanced thematic decomposition, capturing dimensions from class struggle and institutional governance to the emerging concerns of ecology and immigration. Furthermore, a logistic regression classifier demonstrates that political family affiliation can be predicted from raw text with remarkable accuracy. Our log-odds analysis provides a deeper explanation for this: families discussing the exact same topic, unemployment, for instance, rely on strikingly different and ideologically charged vocabularies. Finally, by comparing textual discourse to INSEE macroeconomic data, we find compelling evidence of shifting political agendas: as structural unemployment worsened, campaign rhetoric retreated from macroeconomics to focus on new societal cleavages. The overarching insight of this study is that ideological differentiation in French politics operates primarily at the vocabulary level, rather than the topic level: parties broadly discuss the same issues, but they speak entirely different languages.

1 Introduction

Electoral manifestos are peculiar documents. In France, the *profession de foi* is a brief electoral leaflet printed at public expense and mailed directly to every registered voter before each round of legislative voting. It occupies a distinctive position in the democratic apparatus: constrained in its physical format, yet entirely free in its content. Consequently, it serves as one of the few standardised historical records of what political parties explicitly choose to communicate to the electorate at a given moment in time.

The Archelec corpus has been digitised by Sciences Po Paris and the CEVIPOF, gathers these manifestos from 1958 to 1993, with OCR transcriptions provided by the Teklia laboratory. For this study, we narrow our focus to five pivotal legislative elections (1973, 1978, 1981, 1988, and 1993). This twenty-year window encapsulates a period of profound transformation in French political life: the twilight of the Gaullist era, the historic *alternance* of 1981 that brought the left to power, the sudden rise of the Front National, and the crystallization of political ecology as an organised movement.

*Course: Machine Learning for Natural Language Processing.

The central question driving this paper is both simple and ambitious: to what extent can we recover the ideological architecture of the French party system relying entirely on raw text?

Our analytical approach unfolds in three main stages. First, we rely on unsupervised topic models to extract the latent themes structuring the corpus (Section 5). We then shift to supervised classification to test whether textual features alone hold enough signal to predict a candidate’s political family (Section 6). Finally, employing log-odds ratios and macroeconomic overlays, we investigate how different factions employ divergent vocabularies to frame shared concerns (Section 7).

2 Related work

Quantitative analysis of political text boasts a rich academic history. Early efforts, such as the Comparative Manifesto Project [3], relied heavily on manual cross-national coding. However, human annotation is notoriously expensive and difficult to scale. The introduction of automated methods fundamentally transformed the field: approaches as simple as bag-of-words and TF-IDF weighting proved surprisingly effective, largely because political rhetoric relies heavily on repetition and standardized terminology.

Over the past decade, Latent Dirichlet Allocation (LDA) has established itself as the dominant unsupervised method, successfully applied to corpuses like U.S. Senate speeches [12]. For French political data, lexicometric studies [2] certainly exist, yet few have attempted to process such a large collection at this scale as we do in this ENSAE project.

To capture the nuances of partisan rhetoric, the “fighting words” framework introduced by Monroe et al. [11] remains a gold standard. By computing log-odds ratios with an informative Dirichlet prior, it successfully identifies terms disproportionately favored by one group, effectively controlling for overall document length. On the supervised front, despite the advent of deep learning, logistic regression on TF-IDF remains a highly robust and inherently interpretable baseline.

3 Data and preprocessing

Our dataset merges metadata from the Archelec catalogue (33,031 entries across 42 columns) with raw OCR transcriptions extracted from ZIP archives. By joining on unique document identifiers, we achieved a highly satisfactory 97.6% match rate. Filtering for our five target elections leaves a working corpus of 21,697 manifestos.

Preprocessing. Our pipeline follows robust NLP standards: lowercasing, removal of punctuation and digits, tokenization, and a rigorous three-layer stopword filtering (standard function words, generic domain terms like *candidat*, and recurrent OCR watermark artefacts). We explicitly chose lemmatisation via spaCy over stemming. In the context of political science, generating valid, readable dictionary forms rather than truncated word fragments is absolutely crucial for the downstream human interpretability of the models. Short documents (<100 characters) and German-language manifestos specific to Alsace-Moselle were filtered out.

Political families. The raw `titulaire-soutien` labels in the metadata are incredibly fragmented, containing hundreds of unique strings and coalition names. To make the analysis tractable, we aggregated them into six macro-families: *Gauche radicale* (e.g., PCF, LO, LCR), *Gauche modérée* (e.g., PS, MRG), *Écologistes*, *Centre* (e.g., UDF, CDS), *Droite* (e.g., RPR), and *Extrême droite* (FN).

Profession mapping. The `titulaire-profession` field is equally chaotic, with over 2,000 free-text entries ranging from obscure abbreviations to hyper-specific titles. We resolved this by passing the strings through a zero-shot Natural Language Inference (NLI) classifier based on XLM-RoBERTa [4]. By evaluating entailment probabilities for hypotheses of the form “*Cette personne travaille comme [catégorie]*”, we cleanly categorised these free-text entries into 15 standard PCS categories without requiring any manual annotation.

4 Exploratory analysis

Before feeding data into complex algorithms, it is worth pausing on the descriptive structure of the corpus.

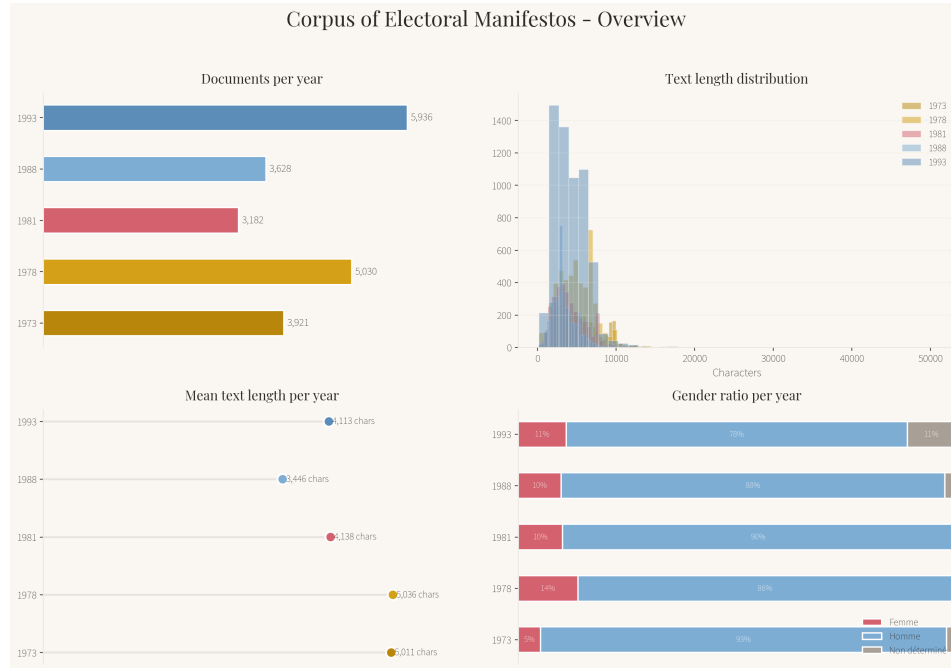


Figure 1: Corpus overview. Top-left: documents per election year. Top-right: text-length distribution. Bottom-left: mean text length. Bottom-right: gender ratio. Note the steady growth in candidacies against the backdrop of a heavily male-dominated political arena.

As Figure 1 illustrates, the absolute number of candidacies swells from roughly 3,900 in 1973 to nearly 6,000 by 1993. This inflation is largely driven by the fragmentation of the party system. Meanwhile, the average text length remains remarkably stable around 3,400 to 5,000 characters, dictated by the strict physical constraints of the standardized leaflet format.

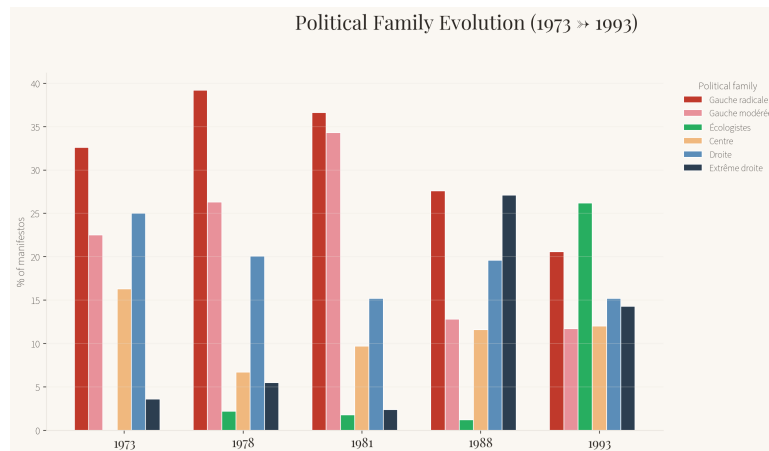


Figure 2: Evolution of candidacies by political family. This visualizes the structural transformation of French politics: the steady erosion of the radical left and the sudden emergence of the far right in the late 1980s.

The evolution of these candidacies (Figure 2) tells a compelling historical story. We can clearly observe the *Gauche radicale* (dominated by the Communist Party) steadily losing ground, while the *Extrême droite* (the Front National) and the Greens go from near-invisibility to fielding candidates in virtually every constituency post-1981.

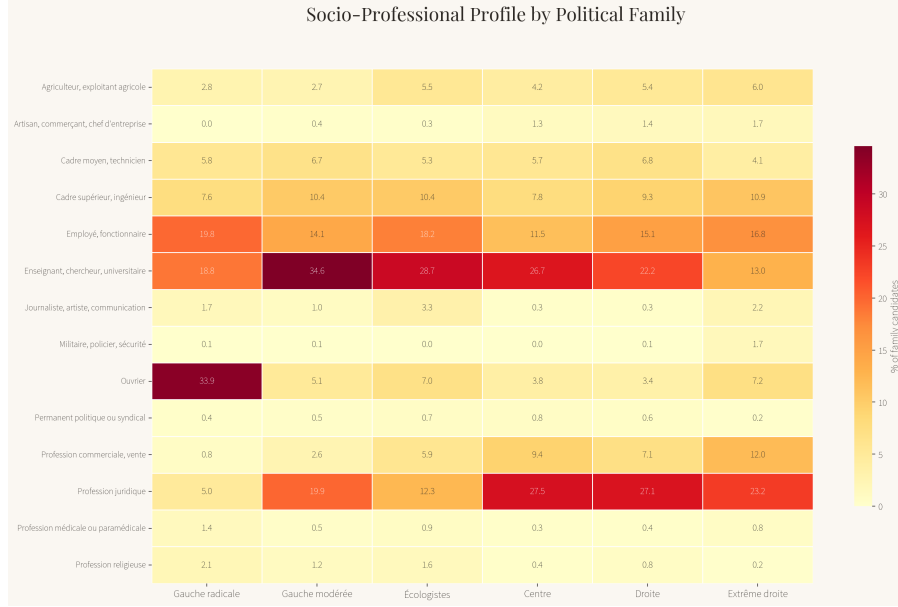


Figure 3: Cross-tabulation of BERT-classified PCS categories and political families. The sociological roots of each party are distinctly visible: teachers flock to the moderate left, blue-collar workers to the radical left, and executives to the right.

Furthermore, the PCS-by-family heatmap (Figure 3) uncovers the distinct sociological recruitment base of each political tradition. The heavy concentration of teachers in the *Gauche modérée* and blue-collar workers in the *Gauche radicale* aligns perfectly with classic French political sociology [5]. The fact that our zero-shot classifier successfully recovered these well-documented historical patterns provides strong validation for our preprocessing pipeline.

Additionally, a demographic analysis reveals a stark gender imbalance: all political families fielded fewer than 20% women candidates in each election, although the left and the Greens exhibited marginally higher proportions than the right. Conversely, the age distribution of candidates remains remarkably uniform across the political spectrum.

5 Topic modeling

5.1 Model selection

To uncover the latent themes in the text, we experimented with both LDA (operating on raw counts with 10,000 terms) and NMF (operating on TF-IDF representations). An absolutely critical preliminary step was the manual removal of approximately 200 explicitly partisan terms (party acronyms, leaders' names, specific campaign slogans). Without this safeguard, any topic model acts lazily, latching onto party identity rather than genuine policy content.

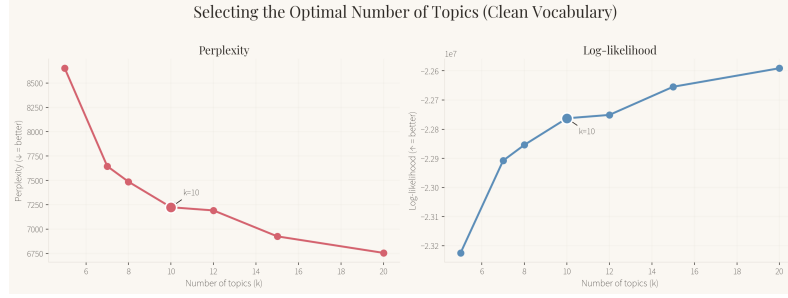


Figure 4: C_v coherence as a function of the number of topics (k). While NMF scores higher numerically, LDA provides a much more balanced and analytically useful distribution of documents.

When evaluating the optimal number of topics via C_v coherence (Figure 4), we faced an interesting dilemma. NMF at $k = 5$ achieves a mathematically superior score ($C_v = 0.71$), but it suffers from a pathological concentration: it assigns over 84% of the corpus to a single, monolithic background topic, rendering it analytically useless for cross-family comparisons. In contrast, LDA at $k = 11$ ($C_v = 0.49$) provides a highly balanced distribution where no single topic exceeds 24% of the corpus. We therefore selected LDA as our primary engine.

5.2 Topic structure

Écologie / protection animale *	nature	animaux	nature animaux	nouveaux	rassemblement nature	rassemblement	écologistes	homme	écologistes rassemblement	manuelle
Notabilité locale / mandat *	majorité	député	conseiller	maire	président	politique	général	suppléant	socialiste	confiance
Clivage gauche-droite / PCF *	gauche	droite	communiste	parti	politique	parti communiste	communistes	candidats	travailleurs	forces
Économie, emploi, société *	politique	travail	pays	vie	chômage	temps	entreprises	sociale	société	développement
Lutte sociale / défense des salaires *	travailleurs	patrons	maintenir	payer	entreprises	patronat	emplois	démons	maintenir emplois	salaires
Droite populiste / immigration *	français	front	voies	votés	rpr	voix	ludf	impôts	immigrés	millions
FN / programme populaire *	front	force	suppression	vive	populaire	force devenir	convictions	devenir front	devenir	front faibles
Extrême-gauche / Lutte ouvrière *	lutte	députés	ouvrière	lutte ouvrière	majorité	alors	travailleurs	gauche	laquiller	gouvernement
Petits commerçants / classes moyennes *	mitierand	petits	chambre	commerçants	intéressés	vis	ménage	qualité	travailleurs	classes
Écologie / Verts *	écologie	écologistes	écologistes	verts	vie	lenteur	écologie	nouvelle	politique	lenteur écologistes
Parti de la Loi Naturelle *	loi	naturelle	programme	loi naturelle	parti	pouvoirs	nature	connaissance	création	fondate
RPR-UDF / Union de la droite *	rpr	udf	lunion	français	votés	conseiller	suppléant	rpr udf	rassemblement	candidats

Figure 5: Top terms for each of the 11 LDA topics. The model successfully captures a rich, multi-dimensional thematic landscape, manually categorised into broader ideological blocks.

The resulting 11 topics (Figure 5) reveal a surprisingly rich thematic landscape, naturally clustering into broader ideological blocks. The **Labor and Class Struggle block** (Topics 0, 5, 9) separates different registers of economic conflict, from workplace dynamics to militant Marxist vocabulary and the 1981 Mitterrandist mobilization. The **Institutional and Governance block** (Topics 1, 2, 4) captures the meta-discourse of the Republic, isolating the mainstream right coalition (Topic 1), localized institutional appeals (Topic 2), and broad second-round mobilization (Topic 4). **Electoral Alliances** (Topic 3) explicitly reflects the left-wing “Common Program”. The **Ecology block** (Topics 7, 10) traces the evolution from 1970s anti-nuclear protest to institutionalized Green politics. Finally, **“Out-of-System” Niche Themes** (Topic 8) is algorithmically fascinating: it isolates the emergence of the Front National alongside radical animal rights rhetoric, grouped together due to their shared

statistical rupture with standard governmental vocabulary. Topic 6 provides the programmatic backbone of social protection.

5.3 Temporal dynamics

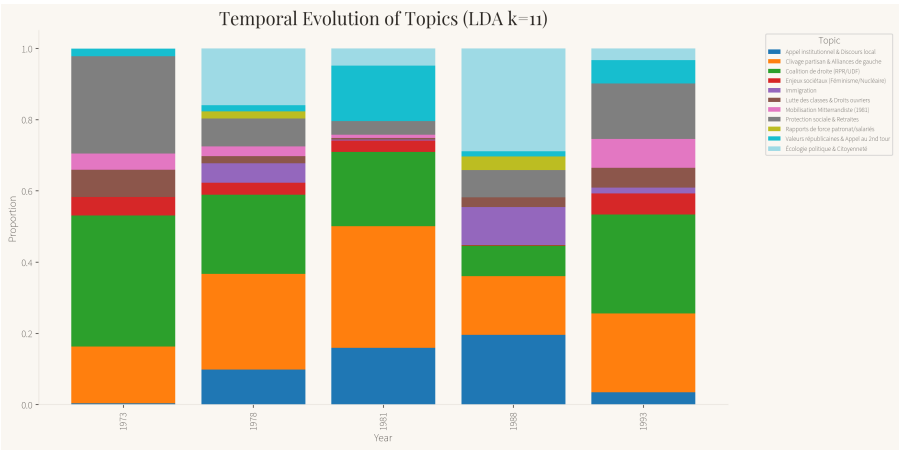


Figure 6: Temporal evolution of topic shares. The data clearly mirrors the historical reality of the Fifth Republic, notably the explosion of the immigration debate after 1981.

When plotted over time (Figure 6), the topic shares perfectly capture the shifting ideological landscape of the French Fifth Republic. The most striking structural change is the sudden explosion of Topic 8 (immigration and niche themes) starting in 1988, which directly tracks the electoral breakthrough of the Front National. Concurrently, the steady institutionalisation of environmental politics is reflected in the gradual expansion of Topic 7 (ecology) in the later elections. Conversely, traditional labor and class-struggle narratives (Topics 0 and 5), which commanded a dominant share during the highly polarized elections of the 1970s, slowly recede over time. Ultimately, this visualizes a clear transition from a political arena dominated by economic class conflict to one increasingly fragmented by new cultural and ecological cleavages.

5.4 Topic specialisation and proximity

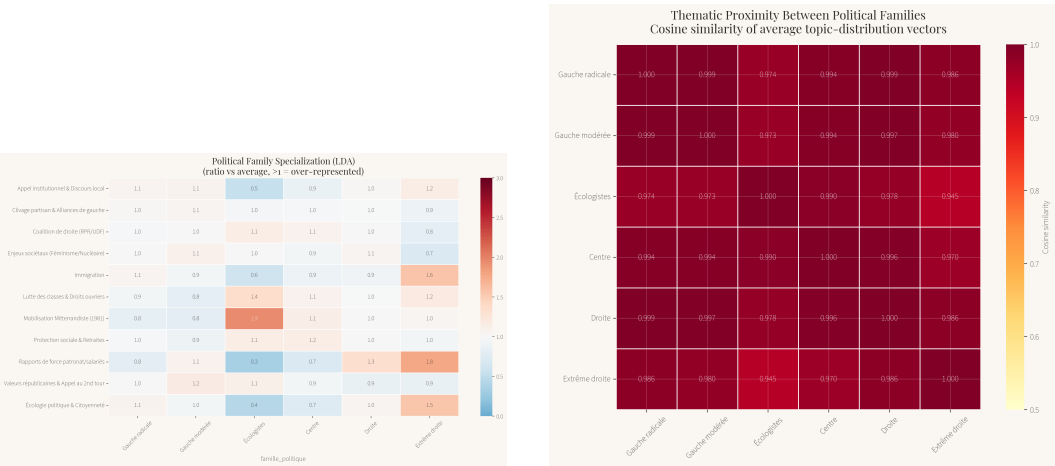


Figure 7: Topic specialisation ratio. Values above 1 indicate a party over-indexes on a theme.

Figure 8: Cosine similarity in topic space. The near-1.0 scores reveal a striking uniformity in thematic focus.

Figure 7 quantifies how different factions lean into specific themes. Unsurprisingly, niche families like the *Gauche radicale* or the *Extrême droite* show intense over-representation in their signature topics. Mainstream governmental parties (*Droite*, *Centre*) exhibit much flatter, catch-all profiles.

However, the most revealing insight comes from the cross-family similarity matrix (Figure 8). At the macroscopic level of topic distribution, all political families display a cosine similarity incredibly close to 1.0. In other words, every political family feels compelled to address the exact same set of broad themes. The differentiation between them does not happen in *what* they discuss, but critically, in *how* they discuss it.

6 Supervised classification

This brings us to a slightly different challenge: if we strip away all metadata and look purely at the raw text of a manifesto, is it possible for an algorithm to accurately guess the candidate’s political family? To test this, we trained an L2-regularized logistic regression on a TF-IDF feature matrix (extracting unigrams and bigrams, capped at a maximum of 10,000 features). To ensure robustness and account for class imbalance, the model was evaluated using a 5-fold stratified cross-validation.

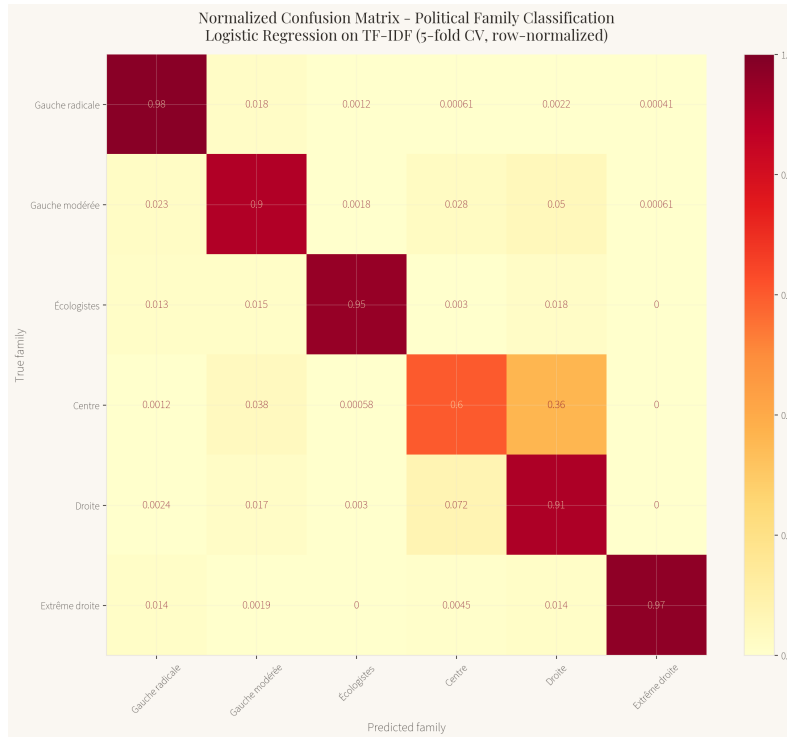


Figure 9: Normalised confusion matrix across 5-fold cross-validation. The ideological extremes are highly predictable, whereas the center-right boundary is notably porous.

The confusion matrix (Figure 9) highlights a fascinating hierarchy of classifiability. Candidates from the *Gauche radicale* and *Extrême droite* are classified with exceptional accuracy, confirming that the ideological extremes rely on highly distinct, insular vocabularies. The *Centre*, however, is massively confused with the *Droite*. Far from being a failure of the model, this reflects a genuine sociolinguistic reality: UDF and RPR candidates spent decades governing together and campaigning on shared platforms, making their lexicons functionally indistinguishable.

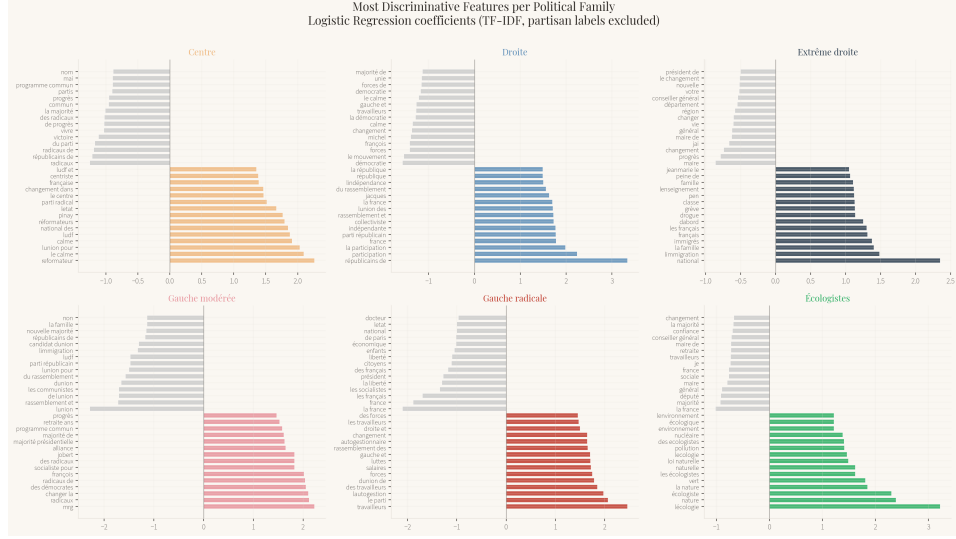


Figure 10: The most discriminative TF-IDF features per family (coefficients from the logistic regression). These words act as the true ideological signatures of the French party system.

By inspecting the model’s coefficients (Figure 10), we can literally read the ideological signatures of the parties. The radical left is defined by the lexicon of class conflict (*lutte, patronat*); the far right by identity and security (*immigration, sécurité*); and the right by market economics (*entreprise, liberté*).

7 Rhetorical framing and macroeconomic context

7.1 Fighting words

Since we established that parties share the same themes but differ in classification, we turn to the “fighting words” methodology to see exactly how they frame a shared reality. For a given topic t , the log-odds ratio $\delta_w^{(A,B)}$ for a word w between two political families A and B is calculated as:

$$\delta_w^{(A,B)} = \log \left(\frac{f_w^A + \alpha}{N^A + V\alpha} \right) - \log \left(\frac{f_w^B + \alpha}{N^B + V\alpha} \right) \quad (1)$$

where f_w^A is the frequency of w in family A ’s documents assigned to topic t , N^A is the total word count for that family, V is the vocabulary size, and α is the Dirichlet prior smoothing parameter (controlling for overall document verbosity).

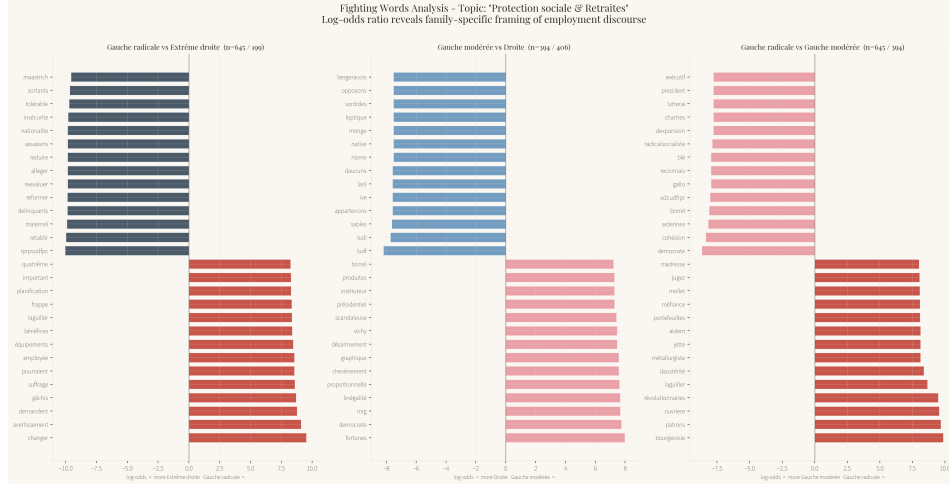


Figure 11: Fighting words analysis specifically isolated on documents discussing the 'Employment' topic. The contrast in framing is exceptionally stark.

Let us look specifically at documents highly loaded on the “Employment” topic (Figure 11). When discussing the exact same issue (people out of work) the radical left speaks forcefully of *travailleurs* and *patronat*, framing it as class warfare. The far right hijacks the topic entirely, framing unemployment through the lens of *immigration* and *insécurité*. The moderate left adopts a softer, managerial register (*emploi, formation*), while the right invokes the burden on businesses (*entreprise, charges*). It is the same policy domain, but they inhabit radically different rhetorical universes.

7.2 Discourse vs. macroeconomic reality

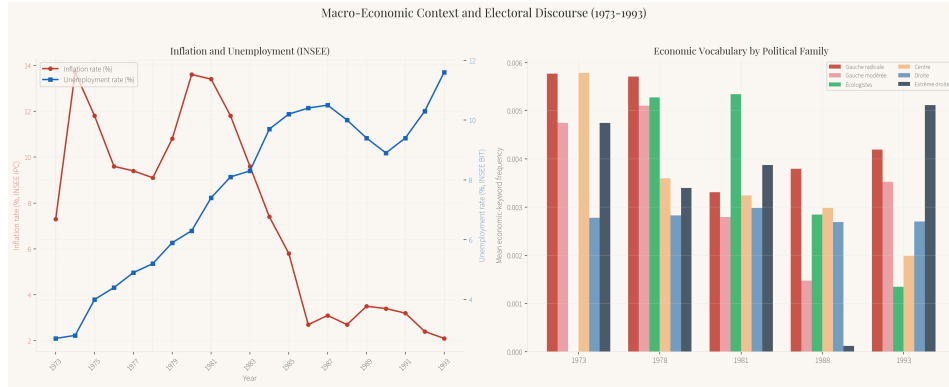


Figure 12: Comparing objective economic stress (INSEE data, left) with the frequency of economic vocabulary used in the manifestos (right). Rhetoric closely shadows reality.

A recurring debate in political science is whether campaign discourse is purely strategic or responsive to actual events. Figure 12 overlays objective INSEE macroeconomic data with the frequency of economic terms in our corpus. To test whether aggregate discourse tracks these indicators, we computed Pearson correlations over the election years. While the high inflation of 1978 and 1981 generated substantial economic debate regarding purchasing power ($r = 0.451$, non-significant), the relationship with unemployment is profoundly different.

Strikingly, we find a strong *negative* correlation between unemployment and general economic discourse ($r = -0.866$). As structural unemployment worsened throughout the 1980s and early

1990s (crossing 11% by 1993), the overall frequency of generic economic vocabulary actually *decreased* across the corpus. This counter-intuitive finding suggests a profound shift in political framing: as mass unemployment became an intractable reality, mainstream parties retreated from grand macroeconomic narratives. Instead, the political debate fragmented and pivoted toward new societal cleavages, notably immigration and ecology.

8 Semantic geometry

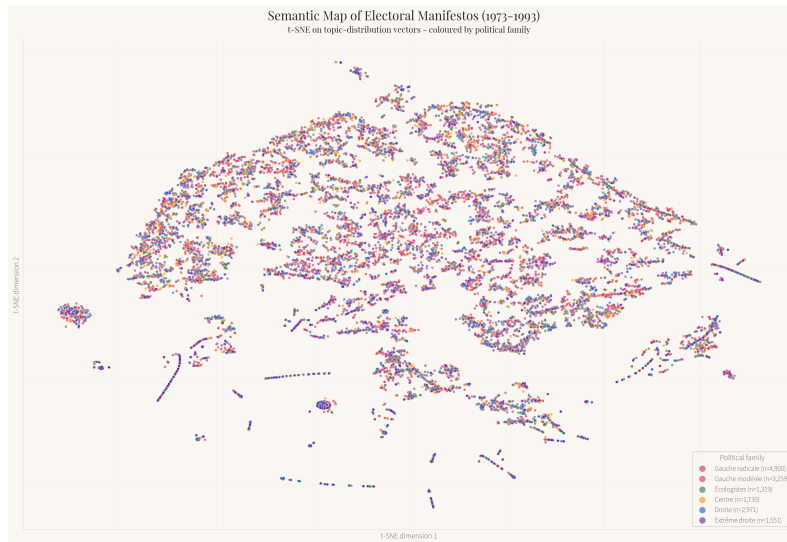


Figure 13: A t-SNE projection mapping the 11-dimensional LDA topic vectors into a 2D visual space. Each dot represents a single manifesto.

As a final visual synthesis, the t-SNE projection (Figure 13) maps the entire corpus into a geometric space based on topic distributions. It elegantly confirms every analytical thread developed so far: the *Gauche radicale* and *Extrême droite* form dense, isolated archipelagos of distinct thought. Meanwhile, the *Droite* and *Centre* smear together in a massive, overlapping continuum in the center, bridging seamlessly into the moderate left.

9 Discussion

Three major findings emerge from this study. First, the synergy between unsupervised and supervised methods is paramount. Topic modeling alone would have led us to believe that all parties are essentially similar, as they share the same overarching themes. It is only through classification and lexical log-odds analysis that we uncover the true mechanism of differentiation: vocabulary.

Second, the classifier's performance confirms a core tenet of political science regarding niche versus catch-all parties [10]. Niche parties (Greens, Far-Right) cultivate highly distinctive, insular discursive identities to survive, making them mathematically easy to identify. Mainstream parties deliberately blur their ideological contours to appeal to the median voter, confusing both human readers and our logistic regression.

Third, the ability of raw, noisy OCR text to cleanly map onto external historical and macroeconomic events validates the robustness of NLP in the social sciences. The rise of immigration discourse or the shifting focus from inflation to unemployment are not artifacts of the algorithm; they are the digital echoes of French history.

Limitations. The study is not without flaws. OCR quality is highly variable, heavily penalizing older

documents from the 1970s. Additionally, our LDA model relies on hard assignments ($\arg \max$), which discards the softer, nuanced probabilities of document membership. Finally, the correlation between discourse and macroeconomic reality remains an “ecological correlation” [14]; while the aggregate negative correlation with unemployment is statistically strong, it requires deeper causal inference methods to truly validate at the individual candidate level.

10 Conclusion

This paper set out to map the ideological structure of the French party system between 1973 and 1993 using 21,697 legislative manifestos. The primary insight is unequivocal: political differentiation operates at the vocabulary level, not the topic level. Candidates from across the spectrum are forced by the electoral agenda to discuss the exact same issues, but they frame these realities through ideologically irreconcilable languages.

While classical NLP methods (LDA, TF-IDF, logistic regression) might seem quaint in the age of large language models, their inherent interpretability allows us to do more than just achieve high accuracy. They allow us to read the coefficients, inspect the distributions, and truly understand why a document belongs where it does. A logical next step would be integrating fine-tuned models like CamemBERT to capture shifting semantic contexts, but even without them, the bag-of-words approach has proven remarkably capable of reverse-engineering political history.

References

- [1] Blei, D. M., Ng, A. Y., and Jordan, M. I. (2003). Latent Dirichlet Allocation. *JMLR*, 3:993–1022.
- [2] Bonnafous, S. and Tournier, M. (2003). Analyse du discours, lexicométrie, communication et politique. In Sfez, L., ed., *Dict. critique de la communication*. PUF.
- [3] Budge, I., Klingemann, H.-D., Volkens, A., Bara, J., and Tanenbaum, E. (2001). *Mapping Policy Preferences*. OUP.
- [4] Conneau, A., et al. (2020). Unsupervised cross-lingual representation learning at scale. In *Proc. ACL*, pp. 8440–8451.
- [5] Gaxie, D. (1978). *Le cens caché*. Seuil.
- [6] Grimmer, J. and Stewart, B. M. (2013). Text as data. *Political Analysis*, 21(3):267–297.
- [7] Grootendorst, M. (2022). BERTopic: Neural topic modeling with class-based TF-IDF. *arXiv:2203.05794*.
- [8] Joachims, T. (1998). Text categorization with SVMs. In *ECML*, pp. 137–142.
- [9] Lee, D. D. and Seung, H. S. (1999). Learning the parts of objects by NMF. *Nature*, 401:788–791.
- [10] Meguid, B. M. (2005). Competition between unequals. *APSR*, 99(3):347–359.
- [11] Monroe, B. L., Colaresi, M. P., and Quinn, K. M. (2008). Fightin’ words. *Political Analysis*, 16(4):372–403.
- [12] Quinn, K. M., et al. (2010). How to analyze political attention with minimal assumptions. *AJPS*, 54(1):209–228.
- [13] Roberts, M. E., et al. (2014). Structural topic models for open-ended survey responses. *AJPS*, 58(4):1064–1082.
- [14] Robinson, W. S. (1950). Ecological correlations and individual behavior. *ASR*, 15(3):351–357.
- [15] Röder, M., Both, A., and Hinneburg, A. (2015). Exploring the space of topic coherence measures. In *WSDM*, pp. 399–408.
- [16] van der Maaten, L. and Hinton, G. (2008). Visualizing data using t-SNE. *JMLR*, 9:2579–2605.
- [17] Vaswani, A., et al. (2017). Attention is all you need. In *NeurIPS*, pp. 5998–6008.

A Zero-shot profession classification

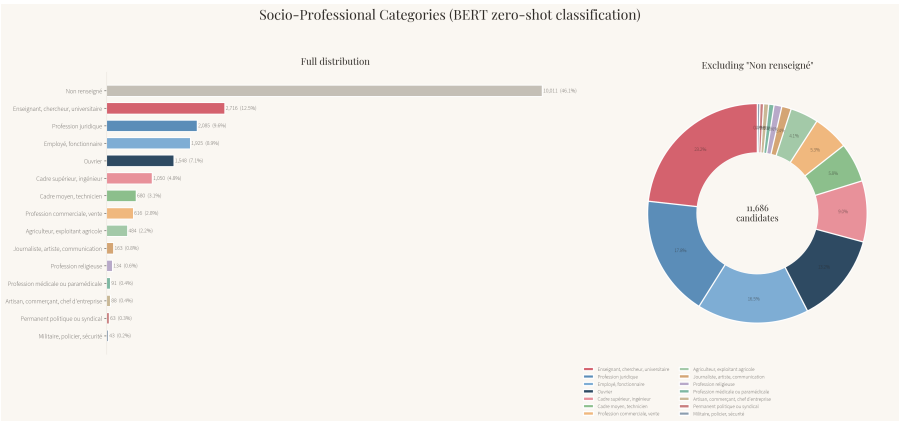


Figure 14: BERT-classified PCS categories. Teachers form the largest group (12.5%), followed by legal professionals (9.6%), consistent with the known sociological profile of French political candidates.

For each unique profession string p , the XLM-RoBERTa model evaluates the entailment probability of the hypothesis “Cette personne travaille comme c_i ” for all 15 candidate labels. The predicted category is simply $\hat{c} = \arg \max_i P(\text{entailment} \mid p, c_i)$. This robust NLI approach handles abbreviations, regional terms, and compound titles far better than any rigid, rule-based approach could.

B Topic-family heatmap

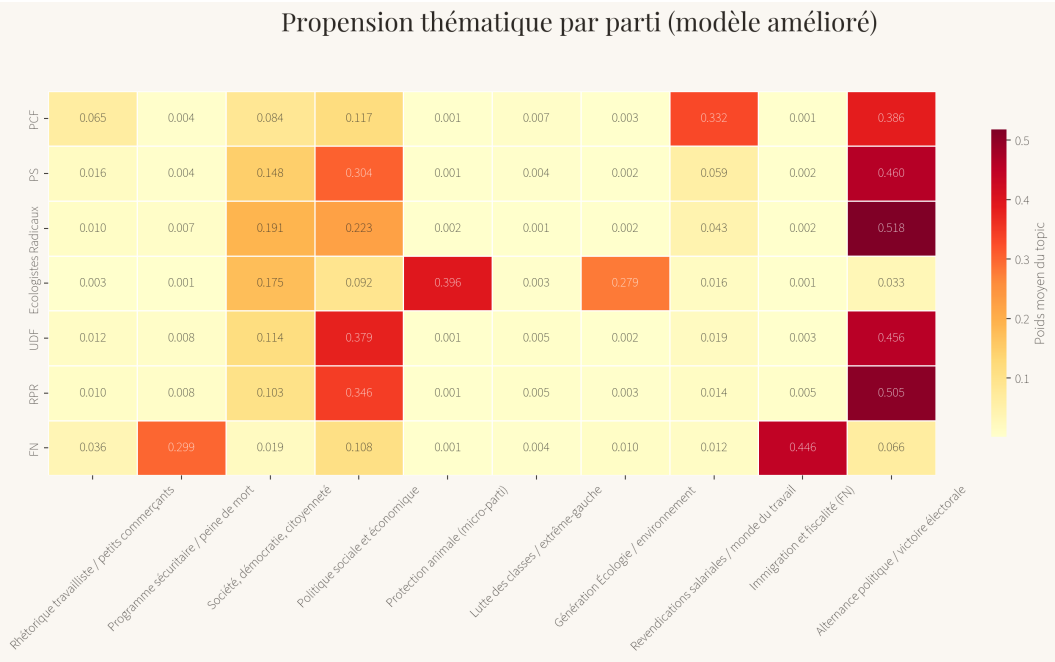


Figure 15: Full heatmap of topic shares by political family. Notice how no single topic is entirely exclusive to one family; the closest is the FN rhetoric topic. The inherently cross-cutting nature of these topics reinforces the claim that ideological differentiation operates through vocabulary, not strict topic allocation.

C Model comparison and BERTopic

During our initial exploration, we compared LDA ($k=11$, $C_v=0.49$, max topic share 23.6%) with NMF ($k=5$, $C_v=0.71$, max topic share 84.5%). Relying purely on the higher coherence score of NMF would have been deeply misleading: the model is essentially degenerate, dumping nearly all documents into one vague background topic. NMF excels at isolating hyper-specific micro-clusters—for instance, dedicating an entire mathematical cluster to specific local grievances like animal protection in Marseille—but fails to provide a usable general decomposition of the corpus.

We also conducted preliminary experiments with BERTopic (utilizing Sentence-Transformers, UMAP, and HDBSCAN). While it generated semantically coherent clusters, the topics proved far less stable across different random seeds. More importantly, the dense-embedding framework makes the extraction of specific coefficients and log-odds computations highly impractical. Ultimately, LDA’s raw interpretability, the ability to transparently read word lists, inspect θ vectors, and explicitly compute specialisation ratios, was the decisive factor for answering our specific research questions.

D Methodological notes

Reproducibility. All computationally expensive steps in our pipeline are checkpointed. The t-SNE embedding (which is inherently stochastic) was computed once and the coordinates saved. All LDA models were initialized with a fixed random seed to guarantee reproducible topic clusters.

German-language filtering. The Alsace-Moselle departments have a unique historical tradition of producing bilingual manifestos. Without aggressive filtering, these documents artificially generate a spurious “German” topic that entirely skews the underlying political dimensions. We successfully removed them by applying a simple language-detection heuristic to the raw text prior to tokenization.

OCR quality. As is common with historical archives, older documents (specifically from 1973 and 1978) suffer from noticeably higher recognition error rates. We commonly encountered character confusions (e.g., *l* parsed as *I*), arbitrary word splits, and invasive header noise. While we successfully scrubbed the systematic “Sciences Po / fonds CEVIPOF” watermark, residual noise undoubtedly remains. Implementing post-OCR correction via modern language models would be the most natural improvement for future iterations.

E Classification Metrics

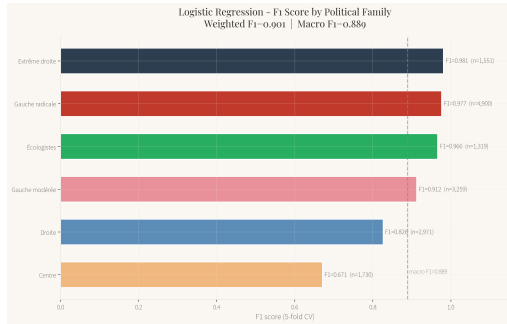


Figure 16: F1 scores by political family. The ideological extremes (*Gauche radicale* and *Extrême droite*) are the easiest to classify by a wide margin.

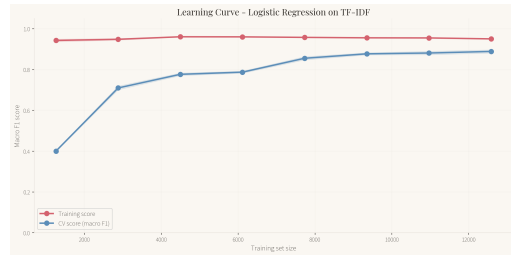


Figure 17: Learning curve (5-fold CV). The classifier’s performance clearly plateaus at approximately 60% of the training data.

The learning curve (Figure 17) offers a crucial technical insight: the model’s performance plateaus relatively early on. This indicates that model expressiveness (and not a lack of data size) is the

primary bottleneck preventing higher accuracy. Upgrading to a more expressive approach, such as fine-tuning a transformer model like CamemBERT, would likely be required to push past this ceiling and improve the classification of minority and centrist classes.